THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A beaker of water is placed on a heater. After some time bubbles form and water level falls. This is: 1

(a) freezing **(b)** melting **(c)** boiling **(d)** condensation

2. A body moving along a circular track at constant speed shows: 1

(a) linear motion **(b)** circular motion **(c)** rest **(d)** no motion

3. Which test would show that a metal rod is a good conductor of heat? 1

(a) wax melts quickly along the rod when one end is heated
(b) it dissolves
(c) it floats
(d) it breaks

4. Why should a lab thermometer not be held by the bulb while reading? 1

(a) it will break **(b)** bulb is sensitive to body heat
(c) it is slippery **(d)** it is cold

5. Which of these is correctly matched? 1

(a) ice – solid **(b)** steam – liquid **(c)** water – gas **(d)** vapour – solid

6. Why is a ruler with clear markings better than one with worn markings? 1

(a) easier to read correctly **(b)** costlier
(c) heavier **(d)** looks better

7. Why are clothes for winter often made of wool? 1

(a) wool is a poor conductor of heat **(b)** wool is shiny
(c) wool dissolves **(d)** wool is hard

8. Water at 20 °C and water at 60 °C are mixed. The mixture could be: 1

(a) 80 °C **(b)** 40 °C **(c)** 70 °C **(d)** 10 °C

9. Which practice increases groundwater? 1

(a) rainwater harvesting **(b)** polluting rivers
(c) cutting trees **(d)** cementing the entire ground

10. A rope of 2.5 m is cut into 5 equal pieces. Each piece in cm is: 1

(a) 25 cm **(b)** 500 cm **(c)** 50 cm **(d)** 5 cm

11. A material is hard, shiny and a good conductor of heat. It is probably a: 1

(a) paper **(b)** wood **(c)** plastic **(d)** metal

12. A hot iron rod and a hot wooden rod, both at 60 °C, are touched. Which feels hotter? 1

(a) iron **(b)** wood **(c)** both equal **(d)** neither

13. Which of the following is correct for 100 °C? 1

(a) dew forms **(b)** ice melts **(c)** water freezes **(d)** water boils

14. The Earth rotating on its axis is: 1

(a) rotational motion **(b)** random motion
(c) no motion **(d)** linear motion

15. Which of the following should be used to hold a hot kettle handle? 1

(a) a thick wooden handle **(b)** an iron rod
(c) a copper rod **(d)** a steel handle

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is a transparent glass not used for bathroom windows? 2

17. Why do we wear several layers instead of one thick layer in winter? 2

18. Explain why ice cubes in a drink melt faster in a hot place. 2

19. Find the time period of a pendulum if it makes 30 oscillations in 60 s. 2

20. A student drops a sugar cube and a salt crystal in water. What is the observation? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Why do we get fresh water from rain though sea water is salty? 3

22. A cyclist covers 15 km. Express in m and cm. 3

23. Why can a stone sink but a stone piece tied to a large piece of thermocol floats? 3

24. Why is a thin cotton cloth preferred in summer? 3

OR

Give three advantages of storing water in dams. 3

25. A thread of 1.1 m wraps round a ball exactly 5 times. Find the circumference in cm. 3

OR

Explain how to find which material is the best conductor of heat among iron, copper and aluminium rods. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain with examples the concept of relative motion for 5

(a) a train passenger and
(b) the Sun as seen from Earth.

OR

Describe how to make a simple filter to separate a mixture of sand and salt in water. 5

27. Explain the idea of thermal equilibrium with the example of mixing hot and cold water. 5

OR

Describe how to prepare drinking water in an emergency using boiling and filtering. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND